

Adjuvant PARP Inhibitors in Patients With High-Risk Early-Stage HER2-Negative Breast Cancer and Germline *BRCA* Mutations: ASCO Hereditary Breast Cancer Guideline Rapid Recommendation Update

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ASCO Rapid Recommendations Updates highlight revisions to select ASCO guideline recommendations as a response to the emergence of new and practice-changing data. The rapid updates are supported by an evidence review and follow the guideline development processes outlined in the ASCO Guideline Methodology Manual. The goal of these articles is to disseminate updated recommendations, in a timely manner, to better inform health practitioners and the public on the best available cancer care options.

BACKGROUND

In 2020, ASCO published a guideline on the management of hereditary breast cancer.¹ On June 3, 2021, the OlympiA phase III, double-blind, randomized trial reported on the efficacy of adjuvant poly(ADP-ribose) polymerase (PARP) inhibitor therapy with olaparib in patients with early-stage, human epidermal growth factor receptor 2 (HER2)-negative breast cancer with high risk of recurrence and germline *BRCA1* or *BRCA2* pathogenic or likely pathogenic variants.² A significant improvement in invasive and distant disease-free survival constituted a strong signal for an update of the 2020 ASCO-ASTRO-SSO guideline recommendation focused specifically on the role of PARP inhibitors in patients with early-stage, HER2-negative breast cancer and germline *BRCA* mutations.

METHODS

A targeted literature search was conducted to identify phase III clinical trials pertaining to the recommendation on PARP inhibitors in this patient population. No additional randomized trials were identified. The original Expert Panel was reconvened to review the evidence from OlympiA and to approve the updated recommendation.

EVIDENCE REVIEW

Tutt et al² reported that, compared with placebo, one year of olaparib following the completion of local treatment and (neo)adjuvant chemotherapy was associated with significantly longer survival free of invasive or distant disease (interim analysis with a median follow-up of 2.5 years). Patients had completed at least six cycles of neoadjuvant or adjuvant chemotherapy; 95% of patients in the trial received

anthracycline- and taxane-based chemotherapy. In the olaparib group, the 3-year invasive disease-free survival was 85.9% versus 77.1% in the placebo group (hazard ratio, 0.58; 99.5% CI, 0.41 to 0.82; $P < .001$). In the olaparib group, the 3-year distant disease-free survival was 87.5% versus 80.4% in the placebo group (hazard ratio, 0.57; 99.5% CI, 0.39 to 0.83; $P < .001$). Anemia was the only grade 3 adverse event reported in $> 5\%$ of patients. In the olaparib group, 8.7% of patients had grade ≥ 3 anemia versus 0.3% of patients in the placebo group. More patients in the olaparib group had at least one blood transfusion with 5.8% versus 0.9% in the placebo group. The occurrence of serious adverse events, including myelodysplastic syndrome and acute leukemia, was not more frequent in the olaparib arm versus placebo.

2020 RECOMMENDATION

Before the publication of the OlympiA data,² the management of hereditary breast cancer joint Panel published this practice recommendation in 2020: For germline *BRCA* mutation carriers, there are insufficient data at this time to recommend a PARP inhibitor for patients with nonmetastatic breast cancer.

2021 UPDATED RECOMMENDATION

The updated recommendation for June 2021 is that for patients with early-stage, HER2-negative breast cancer with high risk of recurrence and germline *BRCA1* or *BRCA2* pathogenic or likely pathogenic variants, one year of adjuvant olaparib should be offered after completion of (neo)adjuvant chemotherapy and local treatment, including radiation. For those who had surgery first, 1 year of adjuvant olaparib should be

ASSOCIATED CONTENT

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offered for patients with triple-negative breast cancer and tumor size > 2 cm or any involved axillary nodes. For those with hormone receptor–positive disease, 1 year of adjuvant olaparib should be offered to those with at least four involved axillary lymph nodes. For patients who had neoadjuvant chemotherapy, 1 year of adjuvant olaparib should be offered to patients with triple-negative breast cancer and any residual cancer; for patients with hormone receptor–positive disease, 1 year of adjuvant olaparib should be offered to patients with residual disease and a clinical stage, pathologic stage, estrogen receptor, and tumor grade score ≥ 3 (Table 1).

TABLE 1. CPS+EG Score Calculation Instructions: Add the Points for Clinical Stage + Pathologic Stage + ER Status + Nuclear Grade to Derive a Sum Between 0 and 6

Stage or Feature	Points
Clinical stage (AJCC staging ^a)	
I	0
IIA	0
IIB	1
IIIA	1
IIIB	2
IIIC	2
Pathologic stage (AJCC staging ^a)	
0	0
I	0
IIA	1
IIB	1
IIIA	1
IIIB	1
IIIC	2
Receptor status	
ER-negative ^b	1
Nuclear grade ^c	
Nuclear grade 3	1

NOTE. Adapted from Tutt et al.² Copyright 2021 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

Abbreviations: AJCC, American Joint Committee on Cancer; CPS+EG score, clinical stage + pathologic stage + ER status + nuclear grade; ER, estrogen receptor.

^aAJCC.³

^bER; definitions for ER negativity, see eligibility criteria section 4.1.4.a.

^cIn the unlikely situation nuclear grade cannot be determined, regular histologic grade should be used; if only Nottingham overall grade is reported, the Nottingham overall grade must be 9 to be scored as 1 point in the Clinical Stage + Pathologic Stage + ER Status + Nuclear Grade score.⁴

DISCUSSION

Although the 3-year estimated overall survival was greater with olaparib, the difference was not statistically significant at the time of this interim analysis at a 2.5-year median follow-up. Of note, postneoadjuvant capecitabine was not permitted in the OlympiA trial because this therapy was not the standard of care when the trial was designed. Thus, the trial cannot inform the relative efficacy of olaparib as compared with capecitabine in this context. Safety and tolerability in OlympiA were manageable, and no significant problems with quality of life (global measurement) were noted at the interim analysis. Further follow-up is needed for myelodysplastic syndrome and acute myelogenous leukemia given that PARP inhibitors are DNA-interacting drugs and have the potential to induce hematologic malignancies. Finally, OlympiA did not assess the effect of olaparib as adjuvant therapy in any hereditary forms of breast cancer other than that associated with germline *BRCA1* or *BRCA2* mutations or assess benefit in patients who lack the high-risk clinical features required for eligibility in this trial.

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GUIDELINE AND CONFLICTS OF INTEREST

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines (“Policy,” found at <http://www.asco.org/rwc>). All members of the Expert Panel

completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

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EDITOR'S NOTE

This ASCO Clinical Practice Guideline Recommendation Update provides a recommendation update, with review and analysis of the relevant literature for the recommendation. Additional information, including links to patient information at www.cancer.net, is available at www.asco.org/breast-cancer-guidelines.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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